

Database Management Systems (DBMS)

PRACTICAL LABORATORY EXAMINATION

1 HOUR

TIME ALLOWED

Scenario: You are appointed as a database administrator for a **Hospital Management System**. Write and execute standard SQL statements in MySQL / phpMyAdmin to fulfill the tasks across all 5 parts below.

PART 1 Data Definition Language (DDL)

1. Create a new database named HospitalDB.
2. Create two relational tables with appropriate primary keys, foreign keys, and constraints:

Patients: `PatientID` (INT, Primary Key), `FullName` (VARCHAR 100), `Email` (VARCHAR 100, Unique), `DateOfBirth` (DATE)

Appointments: `AppointmentID` (INT, Primary Key), `PatientID` (INT, Foreign Key referencing Patients), `DoctorName` (VARCHAR 100), `AppointmentDate` (DATE), `Fee` (DECIMAL 10,2)

3. Alter the Patients table structure to add a new column named BloodGroup (VARCHAR 5).

PART 2 Data Insertion (Data Population)

1. Insert at least **3 patient records** into the Patients table with complete attributes.
2. Insert at least **4 appointment records** into the Appointments table (ensure at least one patient has multiple booked appointments).

PART 3 Data Updating (Data Modification)

1. Update the BloodGroup of the patient with PatientID = 1 to 'O+'.
2. Increase the Fee by 15% for all appointments scheduled after 2024-01-01.

PART 4 Data Deletion & Joins

1. Delete a specific record from the Appointments table where AppointmentID = 2.
2. Write an INNER JOIN query combining Patients and Appointments to retrieve each patient's FullName, DoctorName, AppointmentDate, and Fee.

PART 5 Advanced Querying & Table Truncation

1. Write a query using GROUP BY and SUM() to display total fees collected per patient (showing PatientID, FullName, and total revenue TotalFeeSpent).
2. Execute a DDL statement to truncate the Appointments table. Include a brief SQL comment explaining the functional difference between DELETE FROM Appointments and TRUNCATE TABLE Appointments.